INCH-POUND

MIL-DTL-32832
9 October 2026

DETAIL SPECIFICATION

ROPE, FIBER ASSEMBLY, INSERTION/EXTRACTION, IMPROVED

This document is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers the technical requirements for rope, fiber assembly insertion/extraction, improved.

1.2 Classification. The rope is available in the following types and classes.

1.2.1 Types.

Type I - Fiber, Staple
Type II- Filament, Textured

1.2.2 Classes.

Class I - 60 feet long (+10%, -0%)
Class II - 90 feet long (+10%, -0%)
Class III - 120 feet long (+10%, -0%)

2. APPLICABLE PUBLICATIONS

2.1 General. The publications listed in this section are specified in sections 3 and 4 of this specification. This section does not include publications cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, users of this document are cautioned that they must meet all specified requirements of publications cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be emailed to trpsptctstandardiz@dla.mil. Since contact information can change, verify the currency of this address information using ASSIST at https://assist.dla.mil.

AMSC N/A FSC 4020

DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:48Z
Check the source to verify that this is the current version before use.

2.2 Government publications.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS.

MIL-DTL-32075 - Label: For Clothing, Equipage, and Tentage (General Use)

(Copies of this publication are available online at https://quicksearch.dla.mil).

2.2.2 Other Government publications. The following other Government publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

U.S. ARMY, DEVCOM – SOLDIER CENTER

11-1-3957 - Fiber Rope Assembly, Insertion/Extraction 1/
1/ This is commonly referred to as (FRIES).

(Comments, suggestions, or questions on this document should be addressed to:
Department of the U.S. Army, DEVCOM, Soldier Center, ATTN: FCDD-SCD-PEC,
10 General Greene Avenue, Natick, MA 01760-5019)

(Copies of specifications, standards, drawings and publications required by contractors in connection with specification procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.3 Non-Government publications. The following publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

CORDAGE INSTITUTE (CI)

CI 1500B	Test Methods for Fiber Rope Performance Properties
CI 1501	Hardness Test Method 3 and 8-Strand Fiber Rope

(Copies of these publications are available online at https://www.ropecord.com).

PARACHUTE INDUSTRY ASSOCIATION (PIA)

PIA-TM-6004	Length Per Unit Weight; Cordage

(Copies of this publication are available online at https://pia.com).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:48Z
Check the source to verify that this is the current version before use.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP9 - Evaluation Procedure for Visual Assessment of Color Difference of Textiles
AATCC TM20/TM20A - Test Method for Fiber Analysis: Qualitative and Quantitative

(Copies of these publications are available online at https://aatcc.org).

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this publication are available online at https://asq.org).

SAE INTERNATIONAL

SAE-AMS-STD-595 - Colors Used in Government Procurement

(Copies of this publication are available online at https://www.sae.org).

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Conformance inspection. When specified, a sample shall be subjected to inspection in accordance with 4.4.2.

3.3 Standard sample. Unless otherwise indicated, the finished rope shall match the standard sample for shade and appearance, and shall, be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.2).

3.4 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements and promotes economically advantageous life cycle costs.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:48Z
Check the source to verify that this is the current version before use.

3.5 Design. The end item consists of a fiber rope assembly consisting of eight (8) strands of polyethylene terephthalate type polyester with Type I Fiber, Staple or Type II Filament, Textured wrap plaited together as the main rope with nylon extraction and safety lines braided within the main rope's construction in accordance with drawing 11-1-3957. The main rope has an eye splice on one end and is seared and whipped at the other.

3.6 Material.

3.6.1 Main rope. The main rope shall be fabricated from virgin continuous filament, heat and light resistant polyethylene terephthalate type polyester with option of Type I Staple or Type II Textured wrap. The rope shall be dyed green to approximate any color in the range indicated by color chips 34052, 34079, 34086, 34092, 3~096. 34097, 34098. 34102, 34108, and 34128 of SAE-AMS-STD-595. Testing shall be as specified in 4.4.2.2.

3.6.2 White extraction loops rope. The extraction loops rope shall be fabricated from bright, virgin, continuous filament, heat and resistant nylon of at least 6 denier size, having a minimum tenacity of 8.5 grams per denier. The nylon shall be a long chain polymer made of hexamethylene diamine and adipic acid (Nylon 6,6), or a long chain polymer of epsilon amino caproic acid (Nylon 6). Mixtures of any type shall be prohibited in any one rope. Testing shall be as specified in 4.4.2.2.

3.6.3 Safety line rope. The safety line rope shall be fabricated from continuous filament, U.V. stabilized nylon having a minimum tenacity of 6.5 grams per denier. Mixture of nylon types shall be prohibited in any one rope. The rope shall be dyed black to approximate any color in the range indicated by color chips 27040, 37038, and 37056 of SAE-AMS-STD-595. Testing shall be as specified in 4.4.2.2.

3.7 Color. The color shall be as specified in 3.6.1-3.6.3 or as specified in the contract or procurement documents. The finished fiber rope assembly shall match the standard sample for shade and appearance and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced.

3.7.1 Visual shade matching. The color and appearance shall match the standard sample (see 6.4) when tested as specified in 4.5.1.

3.7.2 Care, maintenance and storage. To ensure the durability and structural integrity of your ropes, follow these essential maintenance procedures. Although the ropes are built to last, they are not immune to stains or damage from improper care.

THIS SPACE INTENTIONALLY LEFT BLANK

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:48Z
Check the source to verify that this is the current version before use.

Care Step	Instructions
Basic Cleaning	Keep cleaning to a minimum. Use a soft, non-abrasive brush to remove dirt.
Removing Oil/Grease	Use warm water mixed with a mild detergent.
Saltwater Contact	You must rinse the rope thoroughly with fresh, clean water within 24 hours.
Proper Storage	Always allow ropes to air-dry completely before storing to prevent damage.

3.8 Physical requirements. The finished rope assembly, consisting of the main rope, extraction loops rope, and safety line rope shall conform to the physical properties listed below when tested as specified in 4.4.6.

TABLE I. Physical requirements for Type I and Type II.

Characteristic	Main rope	Extraction loops rope	Safety line rope
Diameter in inches 1/	1-3/4	9/16	9/16
Circumference in inches at load "P"	5-1/2 ± 1/4	1-3/4 ± 3/16	1-3/4 ± 3/16
Load "P"" (pounds) 2 /	610	65	65
Breaking strength Minimum (pounds)	35,000	9,000	9,000
Hardness Minimum Maximum (pounds)	325 450	4 12	30 60
Feet per pound at load "P"	1.23 ± 5	10.79 ± 5	10.34 ± 5
Elongation Maximum (percent)	59	--	--
Loss in breaking strength after heat aging, not exceeding (percent) of initial breaking strength	10	10	10
Moisture content Maximum (percent)	5	5	5
Extractable matter Maximum (percent)	4	4	4

1/ Not a specification requirement; included for information only.
2/ The load "P" (pounds) shall be equal to 200 times the square of the nominal diameter of the rope in inches ($P=200D^2$).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:48Z
Check the source to verify that this is the current version before use.

3.9 Label and identification. The label shall be in accordance with MIL-DTL-32075 Type VI and Class 11. The contents, size of characters of the inscription and format of Class 11 labels and location of the label shall be as specified in contract or procurement documents (see 6.2.).

3.10 Construction.

3.10.1 Main rope construction. The rope shall be constructed of eight (8) strands arranged in four (4) pairs, where one (1) individual strand shall be laid adjacent to the second strand in each pair and shall conform to the requirements specified herein and in Table I. Individual strands shall be made of one size of singles yarn and shall have an equal number of yarns. The yarn shall be 40000 to 50000 denier and shall be "S" and "Z" twist. The rope shall be constructed without knots or splices in the strands of the rope. However, knots will be permitted in the singles yarn. In preparation of the constituent rope strands, the pattern "SZ" shall be employed in the twisting of the yarns and the individual strands of the two (2) pairs while the components of the remaining pairs shall be twisted in the "ZS" pattern. The finished rope shall be so constructed that in the interweaving procedure, pairs of strands of the former structure shall be twisted in the "Z" direction, while alternating pairs of the latter shall be twisted simultaneously in the "S" direction. Heat setting of the rope or any portion of its components shall not be permitted.

3.10.1.1 Main rope construction of assembly. The main rope shall have an eye splice on one end with a minimum of 4-inches and a maximum of 6-inches inside length. The eye splice shall be made using a minimum of three (3) full tucks and two (2) half tucks. The end shall be seared and whipped.

3.10.2 White extraction loops rope construction. The rope shall be constructed of eight (8) strands arranged in four (4) pairs, where one individual strand shall be laid adjacent to the second strand in each pair and shall conform to the requirement specified herein and in Table I. Individual strands shall be made of one size of balanced 3-ply yarns and shall have equal numbers of plied yarns. The singles yarn shall be 4000 to 10000 denier. The rope shall be constructed without knots or splices in the singles yarn, 3-ply yarns, strands, or rope. Knots in the single filaments are acceptable. In the preparation of the constituent rope strands, the pattern "S2S" twist shall be employed in the twisting of the singles yarns, the 3-ply yarns and the individual strands of two (2) pairs, while the components of the remaining pairs shall be twisted in the "ZSZ" pattern. The finished rope shall be so constructed that in the interweaving procedure, pairs of strands of the former structure shall be twisted in the "Z" direction, while alternating pairs of the latter structure shall be twisted simultaneously in the "S" direction. The finished rope shall be heat set. Dry heat setting is prohibited.

3.10.2.1 White extraction loops rope construction of assembly. The white nylon extraction loops rope shall be passed through the bottom of the first loop and then enter again through the center of the main rope three picks up from the bottom of the first loop to form the first loop. Each white loop shall consist of 10 inches (±1 inch) of exposed rope. After the white rope has been passed back through the center of the main rope forming a loop, it shall back through the center of the main rope one pick up and then passed through the center of itself where it last

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:48Z
Check the source to verify that this is the current version before use.

entered the main rope. The white rope will then be passed through the center of the main rope two picks up from the top of the last loop, then passed back through the center of itself, and passed back through the center of the main roe one pick up from where it had last passed through itself, forming the bottom of the second loop. The same procedure shall be followed to make the second and third white loops. After making the third loop, the white nylon rope shall be passed through the center of the main rope three and one-half picks up from the top of the third loop and then eye spliced around the outer half of the main rope. Each eye splice in the white nylon rope shall consist of four full and two half tucks.

3.10.3 Safety line rope construction. The rope shall be constructed of eight (8) strands arranged in four (4) pairs, where one individual strand shall be laid adjacent to the second strand in each pair and shall conform to the requirements specified herein and in Table I. Individual strands shall be made of one size of balanced 3-ply yarns and shall have equal numbers of plied yarns. The singles yarn shall be 4000 to 10000 denier. The rope shall be constructed without knots or splices in the singles yarn, 3-ply yarns, strands, or rope. Knots in the single filaments are acceptable. In the preparation of the constituent rope strands, the pattern" SZS" twist shall be employed in the twisting of the singles yarns, the 3-ply yarns and the individual strands of two (2) pairs, while the components of the remaining pairs shall be twisted in the "ZSZ" pattern. The finished rope shall be so constructed that in the interweaving procedure, pairs of strands of the former structure shall be twisted in the "Z" direction, while alternating pairs of the latter structure shall be twisted simultaneously in the "S" direction. The finished rope shall be heat set. Dry heat setting is prohibited.

3.10.3.1 Safety line rope construction of assembly. The black nylon safety line rope shall pass through the center of the main rope and one-half picks above, and on the same side as, the upper eye splice of the extraction loops rope. The black nylon safety line rope shall then pass through the two (2) securing passes of the extraction loops rope, on the same side of the main rope, then through the center of the main rope, one and one-half picks above the lower eye splice of the extraction loops rope. After passing through the main rope, the safety line rope will then be passed through the securing passes on the opposite side of the first two, and the rope will then be spliced to itself with a short splice between the top eye and the top securing pass. After being spliced together, the safety line rope shall extend out as far as all three (3) extraction loops, when all three (3) extraction loops are fully extended out at the same time, and shall not interfere with the extraction loops.

3.11 Workmanship. The finished rope shall be uniform in quality and shall conform to the quality of product established by this specification. The occurrence of defects, as specified in Table III, shall not exceed the criteria specified in the contract or purchase order.

4. VERIFICATION

4.1 Quality assurance provisions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:48Z
Check the source to verify that this is the current version before use.

4.1.1 Responsibility for inspection. Unless other specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections that are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.2 Responsibility for compliance. All items shall meet all requirements in sections 3 through 5. The inspection set forth in this specification shall become part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.3 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4.1)
b. Conformance inspection (see 4.4.2)

4.3 Inspection conditions. Unless otherwise specified, excluded, amended, modified or qualified in this specification or applicable procurement documents (see 6.2), all inspections shall be performed in accordance with all the requirements of referenced documents.

4.4 Types of inspections.

4.4.1 First article inspection. The first article, submitted in accordance with 3.1, shall be inspected for appearance, color, finished measurements listed in 3.7, 3.7.1 and 3.8, examination for defects listed in Table III, and tested for the characteristics as specified in Table I. If first article is required for the rope, it shall be in accordance with 4.4.2.1, and shall be inspected, examined for appearance, color, and finished defects as specified in Table III and tested for the characteristics as specified in Table I.

4.4.2 Conformance inspection. Conformance inspection shall include the characteristics identified in Table II, defects listed in Table III and the tests of 4.4 through 4.5 as applicable. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 and

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:48Z
Check the source to verify that this is the current version before use.

with acceptable quality limits (AQLs) as specified in the contract and/or procurement documents, except where otherwise indicated (see 6.2).

4.4.2.1 Material inspection. Materials shall be inspected in accordance with all the requirements of referenced documents unless excluded, amended, modified, or qualified in this Specification or applicable purchase document.

4.4.2.2 Material testing. The main rope, extraction loops rope, and safety line rope shall each be tested for the characteristics listed in Table II. The lot size shall be expressed in units of yards of rope. The sample unit shall be 12 yards of rope for the main rope, and 14 yards each for both the extraction loops rope and safety line rope. The inspection level shall be S-1. The lot shall be unacceptable if one or more of the sample units fails to meet any requirement.

TABLE II. Material testing.

Characteristic	Requirement reference	Test method
Material		
Main rope	See 3.10.1 and 3.10.1.1	AATCC TM20/TM20A 1/
Extraction loops rope	See 3.10.2 and 3.10.2.1	AATCC TM20/TM20A 1/
Safety line rope	See 3.10.3 and 3.10.3.1	AATCC TM20/TM20A 1/
Color	See 3.7- 3.7.1	See 4.5.1
Breaking Strength	See Table I	CI 1500B, See 4.4.6.1 2/
Breaking Strength after Heat Aging	See Table I	4/
Hardness	See Table I	CI 1501 2/
Moisture content	See Table I	AATCC 20A
Extractable matter	See Table I	AATCC 20A
Length per pound	See Table I	PIA-TM-6004
Circumference	See Table I	3/

1/ A certificate of compliance shall be submitted.
2/ The length of the rope previously used in the weight determination taped off at each end and used as the test sample.
3/ The circumference shall be measured at the beginning of the breaking strength test with the specimen under the load "P" specified in Table I. A hard fiber shall be passed snugly around the rope and cut where it overlaps. The cut length shall be straightened and measured to the nearest 1/4 inch for the main rope and to the nearest 1/8 inch for the extraction loops rope and safety line rope. This determination shall be repeated at least three times in different positions not less than two turns of rope apart. The average of these determinations shall be the circumference of the rope.
4/ The main rope, extraction loops rope, and safety line ropes shall be heated for 5 days in a convection air oven at 175°F ± 2°F. After aging, the specimens shall be allowed to reach equilibrium under standard conditions before being tested for breaking strength per 4.4.6.1. The percent change in strength shall be calculated based on the average strengths initially and after heat aging.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:48Z
Check the source to verify that this is the current version before use.

4.4.3 End item critical defect examination. Every end item, prior to the visual and dimensional examinations in 4.4.3 and 4.4.4, shall be examined to the critical defects listed in Table III. Any end item found to contain one or more critical defects shall be rejected.

4.4.4 End item visual examination. The finished fiber rope assemblies shall be examined for the defects listed in Table III. The lot size shall be expressed in units of finished rope assemblies. The sample unit shall be one finished assembly. The inspection level shall be S-4, and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 1.5 for major defects and 4.0 for total (major and minor combined) defects. Any critical defect found during sampling shall be cause for rejection of the lot represented by the sample.

TABLE III. Defects.

Examine	Defect	Classification		
		Critical	Major	Minor
Appearance and workmanship	Not constructed as specified	1		
	Any cut	2		
	Chafed or damaged	3		
	Kink	4		
	Knot (singles, ply, strand, or rope)	5		
	Broken or loose ends (singles, ply, strand or rope)	6		
	Bulged strand	7		
	Ends not securely whipped, taped, or heat sealed to prevent fraying or untwisting			201
Color	Other than specified		101	
	Resists and/or spotty dyeing			202
	Uneven			203
Cleanliness	Spot or stain, clearly visible 1/			204
Identification	Omitted, incorrect, illegible, or not as specified			205

1/ At normal inspection distance (approximately 3 feet).

4.4.5 Dimensional examination. The fiber rope assemblies shall be examined for the dimensions specified in the applicable drawing. Any dimension not within the specified tolerance shall be expressed in units of fiber rope assemblies. The sample unit shall be one fiber rope assembly. The inspection level shall be S-3, and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

4.4.6 End item testing. The end item shall be tested as specified in 4.4.6.1. The lot shall be expressed in units of fiber rope assemblies. The sample unit shall be one fiber rope assembly. The inspection level shall be S-1. Any test failure shall be cause for rejection of the lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:48Z
Check the source to verify that this is the current version before use.

4.4.6.1 End item breaking strength and elongation testing. The main rope, extraction loops rope, and safety line ropes of the finished fiber rope assembly shall be tested for breaking strength in accordance with CI 1500B, uncycled -except that the load "P", as determined in Table I, shall be used in lieu of 1 percent of the breaking strength for measurement of "O" in the elongation calculation and the extraction loops rope and safety line rope specimens shall be prepared as follows:

The main rope of the fiber rope assembly shall be cut so that at least 3 feet of clear main rope remains above the top extraction loop. As specified in CI 1500B, uncycled, the cut end shall be eye spliced, and the round metal pins or posts from the testing machine shall hold the specimen at the main rope eye splice and at the loop being tested. No more than one extraction loops rope and one safety line rope shall be tested per each fiber rope assembly. Any failure to meet the requirements in 3.8 shall be classified as a test failure.

4.5 Methods of inspection

4.5.1 Visual shade matching (all Classes). The color and appearance of the rope assembly shall match the standard sample when viewed using the AATCC EP 9, Option C, (see 6.6) with a primary light source simulating the spectral quality of average daylight, CIE Illuminant D65, with a color temperature of 6500 (± 200) Kelvin (K) illumination of 100 (± 20) foot candles. Alternatively, the color and appearance of the rope assembly shall match the standard sample with a primary light source simulating artificial daylight, CIE Illuminant D75, with a color temperature of 7500K (± 200) illumination of 100 (± 20) foot candles in lieu of D65. (See 6.6.1). The rope assembly shall also be a good match to the standard sample with a secondary source simulating the spectral quality of incandescent lamplight, CIE Illuminant A, with a color temperature of 2856K (± 200).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The rope assembly is intended for use during both the infiltration and exfiltration of military personnel from rotary wing aircraft at elevations up to 120 feet above ground level.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:48Z
Check the source to verify that this is the current version before use.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type and Class required (see 1.2).
c. The specific issue of individual documents referenced (see 2.1).
d. When first article is required (see 3.1).
e. Inspection conditions (see 4.3).
f. Performance inspection with acceptance quality limits (AQL) (see 4.4.2).
g. Packaging (see 5.1).

6.3 First article. When a first article inspection is required (see 3.1), it will be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Standard sample. For access to samples and pattern drawings, address the contracting activity issuing the invitation for bids or request for proposal.

6.5 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for that specified requirement.

6.6 Visual shade matching. In 2017, Option A of AATCC Evaluation Procedure 9, Visual Assessment of Color Difference of Textiles was changed to Option C.
NOTE: In case of confusion, the viewing geometry should be "The specimen plane and illumination source will be parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35 (± 5°) angle relative to the horizontal. The observer will view the specimens at a 90° angle, relative to the plane of the specimens".

6.6.1 Lighting sources. In 2022, the U.S. Military (All Services) accepted D65 as the new preferred light source for visual shade matching due to the supply chain issues with D75. DD1222's should reflect which light source is used. If the light source is not indicated on the DD1222, D65 will be assumed by the Government Shade Laboratory.

6.7 Superseding documents. The item specification was previously covered by MIL-F-44422 dated 27 December 1990. This specification is improved and now identifies multiple Types.

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes. The package paragraphs listed in section 4 have been removed and replaced with the specification information located in section 5. NOTE: All Federal Specifications and Department of Defense Standards have been canceled as of 10-29-1999, without any replacements, and have been removed from this document. FED-STD-191 is hereby inactive for new design. FED-STD-595 has been superseded by SAE-AMS-STD-595.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:48Z
Check the source to verify that this is the current version before use.

6.9 Subject term (key word) listing.

Aircraft
Cordage
Equipment, aircraft
Exfiltration
Helicopter
Infiltration

Custodians:
Army - GL
Navy- SH
Air Force- 11

Preparing activity:
DLA-CT

Review activities:
DLA- IS
Navy- MC

(Project: 4020-2026-009)

NOTE: The activities listed above were interested in this document as of date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:48Z
Check the source to verify that this is the current version before use.